

WEEK-6

TASK-1:

create a client-sewer application using Named Pipes (FIFOs). The client sends messages and the Server processes and responds to them. Analyse FIFO behaviour when multiple clients communicate simultaneously.

TERMINAL-1:

```
root@LAPTOP-RJFD8EBT:~# rm -f /tmp/server_fifo /tmp/client_fifo_1 /tmp/client_fifo_2 /tmp/client_fifo_3
root@LAPTOP-RJFD8EBT:~# rm -f server.c client.c server_fifo
root@LAPTOP-RJFD8EBT:~# ls
File1.txt  copyfile.c  process_demo.c  source.txt  task5.c
File2.txt  destination.txt  process_exp.c  task1.c  task5_old
Ossp      file.txt    process_exp.c  task1.c  test.txt
SKILL_Week_4  fork_process  process_status  task2.c  trace.txt
command_executor  fork_process.c  process_status.c.save  task3.c  week3
command_executor.c  information.txt  process_status.c.save  task3.c  week3
copy        ls_grep      producer_consumer.c  task4.c
'copy.c'    ls_grep.c    producer_consumer.c  task4.c
root@LAPTOP-RJFD8EBT:~# nano server.c
root@LAPTOP-RJFD8EBT:~# gcc server.c -o server
root@LAPTOP-RJFD8EBT:~# ls -l server.c server
-rwxr-xr-x 1 root root 16664 Sep 16 18:58 server
-rw-r--r-- 1 root root 2884 Sep 16 18:49 server.c
root@LAPTOP-RJFD8EBT:~# nano client.c
root@LAPTOP-RJFD8EBT:~# gcc client.c -o client
root@LAPTOP-RJFD8EBT:~# ls -l client.c client
-rwxr-xr-x 1 root root 16664 Sep 16 18:51 client
-rw-r--r-- 1 root root 2123 Sep 16 18:50 client.c
root@LAPTOP-RJFD8EBT:~# ./server
=====
FIFO SERVER STARTED
=====
Server FIFO : /tmp/server_fifo
Waiting for clients...

Client FIFO : /tmp/client_fifo_1
Message : Hello Server
Response : Server processed: Hello Server
Response sent successfully.

Client FIFO : /tmp/client_fifo_2
Message : Hello from Client 2
Response : Server processed: Hello from Client 2
Response sent successfully.

Client FIFO : /tmp/client_fifo_2
Message : Hello from Client 2
Response : Server processed: Hello from Client 2
Response sent successfully.

Client FIFO : /tmp/client_fifo_2
Message : Hello from Client 2
Response : Server processed: Hello from Client 2
Response sent successfully.

Client FIFO : /tmp/client_fifo_3
Message : Ossp Practical
Response : Server processed: Ossp Practical
Response sent successfully.

Client FIFO : /tmp/client_fifo_1
Message : Message from Client 1
Response : Server processed: Message from Client 1
Response sent successfully.

Client FIFO : /tmp/client_fifo_2
Message : Message from Client 2
Response : Server processed: Message from Client 2
Response sent successfully.

Client FIFO : /tmp/client_fifo_3
Message : Message from Client 3
Response : Server processed: Message from Client 3
Response sent successfully.
*c
Server stopped.
root@LAPTOP-RJFD8EBT:~# strace -f -e trace=mkfifo,openat,read,write,close,unlink ./server
strace: invalid system call 'mkfifo'
root@LAPTOP-RJFD8EBT:~# ./client 1 "Test FIFO"
Server is not running: No such file or directory
root@LAPTOP-RJFD8EBT:~# rm -f /tmp/server_fifo /tmp/client_fifo_1 /tmp/client_fifo_2 /tmp/client_fifo_3
root@LAPTOP-RJFD8EBT:~#
```

TERMINAL-2:

```
root@LAPTOP-RJFD8EBT:~# ./client 1 "Hello Server"

=====
CLIENT 1
=====
Sent : Hello Server
Received : Server processed: Hello Server
root@LAPTOP-RJFD8EBT:~# ./client 2 "Hello from Client 2"

=====
CLIENT 2
=====
Sent : Hello from Client 2
Received : Server processed: Hello from Client 2
root@LAPTOP-RJFD8EBT:~# ./client 3 "Ossp Practical"

=====
CLIENT 3
=====
Sent : Ossp Practical
Received : Server processed: Ossp Practical
root@LAPTOP-RJFD8EBT:~# ./client 1 "Message from Client 1"

=====
CLIENT 1
=====
Sent : Message from Client 1
Received : Server processed: Message from Client 1
root@LAPTOP-RJFD8EBT:~# ./client 2 "Message from Client 2"

=====
CLIENT 2
=====
Sent : Message from Client 2
Received : Server processed: Message from Client 2
root@LAPTOP-RJFD8EBT:~# ./client 3 "Message from Client 3"

=====
CLIENT 3
=====
Sent : Message from Client 3
Received : Server processed: Message from Client 3
root@LAPTOP-RJFD8EBT:~# ls -l /tmp/server_fifo
prw-r--r-- 1 root root 0 Sep 16 19:07 /tmp/server_fifo
root@LAPTOP-RJFD8EBT:~# ps -f
UID      PID     PPID    C  STIME TTY          TIME CMD
root      583      580    0  18:56 pts/0        00:00:00 -bash
root      727      583    3  19:12 pts/0        00:00:00 ps -f
root@LAPTOP-RJFD8EBT:~# ^[[2~
```

TASK-2:

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSRI. Demonstrate asynchronous event handling and explain the role of signal handlers.

TERMINAL-1:

[illegible]

TERMINAL-2

```

root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 1234
-bash: kill: (1234) - No such process
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# ps -p 1234 -f
UID          PID     PPID  C  STIME TTY          TIME CMD
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 840
root@LAPTOP-RJFD8EBT:~# ^C
root@LAPTOP-RJFD8EBT:~# ^C
root@LAPTOP-RJFD8EBT:~# kill -9 840
root@LAPTOP-RJFD8EBT:~# ps -p 1234
      PID TTY          TIME CMD
root@LAPTOP-RJFD8EBT:~# getpid()
> sigaction()
-bash: syntax error near unexpected token `sigaction'
root@LAPTOP-RJFD8EBT:~# getpid()
> SIGINT
SIGTERM
SIGUSR1
-bash: syntax error near unexpected token `SIGINT'
SIGTERM: command not found
SIGUSR1: command not found
root@LAPTOP-RJFD8EBT:~# sigaction()
> SIGINT
SIGTERM
SIGUSR1
-bash: syntax error near unexpected token `SIGINT'
SIGTERM: command not found
SIGUSR1: command not found
root@LAPTOP-RJFD8EBT:~# kill -SIGUSR1 928
root@LAPTOP-RJFD8EBT:~# ps -p 928 -f
UID          PID     PPID  C  STIME TTY          TIME CMD
root 928      485    0 19:52 pts/2    00:00:00 ./signal_handler
root@LAPTOP-RJFD8EBT:~# kill -9 928
root@LAPTOP-RJFD8EBT:~#
```

